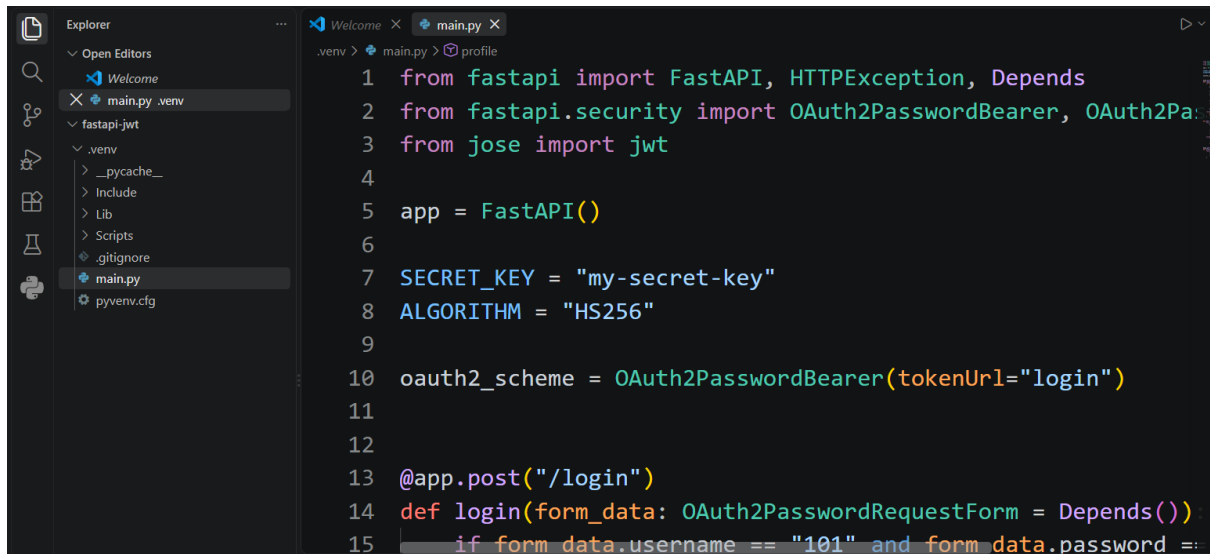


SKILL 6 DBMS



```
1 from fastapi import FastAPI, HTTPException, Depends
2 from fastapi.security import OAuth2PasswordBearer, OAuth2Pa
3 from jose import jwt
4
5 app = FastAPI()
6
7 SECRET_KEY = "my-secret-key"
8 ALGORITHM = "HS256"
9
10 oauth2_scheme = OAuth2PasswordBearer(tokenUrl="login")
11
12
13 @app.post("/login")
14 def login(form_data: OAuth2PasswordRequestForm = Depends())
15     if form_data.username == "101" and form_data.password ==:
```

```
17     payload = {
18         "sub": form_data.username,
19         "role": "student"
20     }
21
22     token = jwt.encode(
23         payload,
24         SECRET_KEY,
25         algorithm=ALGORITHM
26     )
27
28     return {
29         "access_token": token,
30         "token type": "bearer"
31     }
```

```
32
33     raise HTTPException(
34         status_code=401,
35         detail="Invalid username or password"
36     )
37
38
39 def get_current_user(token: str = Depends(oauth2_scheme)):
40     try:
41         payload = jwt.decode(
42             token,
43             SECRET_KEY,
44             algorithms=[ALGORITHM]
45
```

```
47         return payload
48
49     except Exception:
50         raise HTTPException(
51             status_code=401,
52             detail="Invalid or expired token"
53         )
54
55
56 @app.get("/profile")
57 def profile(user=Depends(get_current_user)):
58     return {
59         "message": "Welcome to your profile",
60         "user_id": user["sub"],
```

Authorize 

default ^

POST /login Login v

GET /profile Profile  v

Schemas ^

Body_login_login_post > Expand all object

HTTPValidationError > Expand all object

ValidationError > Expand all object

No parameters

Request body required

grant_type
string | (string | null)
pattern: ^password\$
☐ Send empty value

username * required
string

password * required
string(\$password)

scope
string
☒ Send empty value

client_id
string | (string | null)
☐ Send empty value

client_secret
string | (string | null)(\$password)
☐ Send empty value

Execute

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8080/login' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/x-www-form-urlencoded' \
  -d 'grant_type=password&username=101&password=abc123&scope=&client_id=string&client_secret=*****'
```

Request URL

http://127.0.0.1:8080/login

Server response

Code	Details
200	Response body <pre>{ "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IWRXLCJybzI6IjoiY3R1ZGVudCJ9.NHqwkDLP81Q5BFQ1iKmdoWlOjbLUMdWm4aRQ5MK1zIT", "token_type": "bearer" }</pre> Response headers <pre>content-length: 162 content-type: application/json date: Tue, 15 Sep 2026 05:14:29 GMT server: uvicorn</pre>

Responses

Code	Description
200	Successful Response Media type application/json Controls Accept header: Example Value Schema

Available authorizations

Scopes are used to grant an application different levels of access to data on behalf of the end user. Each API may declare one or more scopes.

API requires the following scopes. Select which ones you want to grant to Swagger UI.

OAuth2PasswordBearer (OAuth2, password)

Authorized

Token URL: login

Flow: password

username: 101

password: *****

Client credentials location: basic

client_secret: *****

[Logout](#) [Close](#)

GET

/profile Profile

Parameters

No parameters

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/profile' \
  -H 'accept: application/json' \
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ2ZmTiOiIyMDE1CjY6Z2xiIjoic3R1ZGVudCJ9.NkpwADLP8lQ5BFQ1iKmdoAvL0jblUMdHm4aRQ6Wk1zI'
```

Request URL

http://127.0.0.1:8000/profile

Server response

Code

Details

200

Response body

```
{
  "message": "Welcome to your profile",
  "user_id": "181",
  "role": "student"
}
```

Download

Response headers

```
content-length: 70
content-type: application/json
date: Tue, 15 Sep 2026 05:16:26 GMT
server: uvicorn
```

Responses

Code

Description

Links

200

Successful Response

No links

Media type

application/json

Controls Accept header.